YEAR 10 PHYSICS, AQA GCSE COMBINED SCIENCE: TRILOGY (HIGHER) AND PHYSICS

# ELECTRIC CIRCUITS: TEACHER GUIDE

Twelve 50-minute lessons, written for students with **no prior knowledge of electricity**. Each lesson has its own booklet in the student workbook.

## HOW THE THREE RESOURCES FIT TOGETHER

| RESOURCE | USED FOR |
|---|---|
| **Student workbook** | One booklet per lesson (12 pt text, wide writing lines), each with its toolkit, key words and quick answers, plus a Unit review booklet with the static electricity page (Triple only) and progress tracker. Every lesson: explanations, worked examples, graded practice, lab records, an exam question and a revision box. It is the students' classwork book and their revision guide. |
| **Answer edition** | Same pages and page numbers as each booklet, with answers in blue. For marking, and for release to students for self-marking if you choose. |
| **Virtual labs, practicals** | Lessons 2–4 use the Virtual Labs worksheet (PhET) for the full investigation; the workbook holds a short record. Lab 3 plots current against p.d., as AQA does. Lessons 5, 7 and 9 are the required practicals: the workbook is the lab sheet. |

## DESIGN PRINCIPLES

- **Concrete before abstract.** Lesson 1 builds real circuits and teaches symbols and drawing rules before any quantity is measured. The voltmeter appears only when students know what p.d. is (Lesson 3).
- **One new idea at a time.** Current, p.d. and resistance get a lesson each, each with a model (bike chain, delivery vans, collisions) and its limitation. Common misconceptions appear as student claims (Mia, Leo, Ava) to correct.
- **Follow along → your turn → faded practice.** Every worked example on a slide is printed blank at the same point in the workbook, tagged with its slide number, so students fill it in while you model it; a parallel 'Your turn' with new numbers sits beside it, and the answers are only in the answer edition. Each Quick check slide has its own box before the Exam corner (after Lessons 2, 5, 7, 10 and 12). Calculation questions go from a full frame (Equation, Substitute, Answer) to no frame to a challenge.
- **Retrieval every lesson.** 'Do now' questions revisit earlier lessons; 'Revise it' gives cover-and-answer questions with quick answers at the back.
- **Equations are still learned.** AQA provides the full Physics Equations Sheet in every paper, confirmed for 2027 and from 2028 onwards. It does not include $R_{total} = R_1 + R_2$, and students who know the equations work faster, so the booklet still teaches recall.

## SEQUENCE

| # | LESSON |
|---|---|
| 1 | **Complete circuits and symbols**<br>What a circuit needs, circuit symbols and drawing rules, measuring current |
| 2 | **Current and charge**<br>Current as a rate of flow of charge, Q = I t, current is not used up |
| 3 | **Potential difference**<br>p.d. as energy per coulomb, voltmeters, E = Q V, p.d.s add in series |
| 4 | **Resistance and V = I R**<br>Resistance, V = I R, ohmic conductors, why heating raises resistance |
| 5 | **Resistance of a wire** REQUIRED PRACTICAL<br>Required practical: how the length of a wire affects its resistance |
| 6 | **Series and parallel circuits**<br>Current, p.d. and resistance rules for series and parallel circuits |
| 7 | **Resistors in series and parallel** REQUIRED PRACTICAL<br>Required practical: measure total resistance of two resistors in series and in parallel |
| 8 | **I–V graphs**<br>I–V graphs of a fixed resistor, filament lamp and diode |
| 9 | **Investigating I–V characteristics** REQUIRED PRACTICAL<br>Required practical: measure the I–V characteristics of a resistor, lamp and diode |
| 10 | **Thermistors, LDRs and sensors**<br>How thermistors and LDRs respond to temperature and light, and sensor circuits |
| 11 | **Power and energy transfer**<br>Power, E = P t, P = V I, P = I² R, energy transfers in appliances |

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| # | LESSON |
|---|---|
| 12 | **Mains, safety and the National Grid**<br>ac and dc, 230 V 50 Hz mains, three-core cable, earthing, the National Grid |

Lesson

# 01 COMPLETE CIRCUITS AND SYMBOLS

AQA Combined 6.2.1.1; Physics 4.2.1.1; 50 minutes

**BIG QUESTION** **Your torch won't switch on. The bulb is fine and the batteries are new. What could be wrong?**

**STUDENTS WILL BE ABLE TO**

- explain why a lamp only lights in a complete loop
- recognise and draw standard circuit symbols
- turn a real circuit into a neat circuit diagram
- connect an ammeter in series to measure current

**WHY THIS LESSON, HERE**

Students start with no formal knowledge. Before any quantity or equation, they need a working idea of a circuit and the vocabulary to draw one. The old Lesson 1 asked for six symbols that were never taught; this lesson teaches them explicitly, from real apparatus to symbol to diagram.

## PREPARATION AND EQUIPMENT

Per pair: 1.5 V cell (or 2 × AA in a holder), lamp in holder, switch, 4 leads, ammeter (0–1 A). One broken torch for the hook (loose battery contact). Optional: a bike or a chain loop to show the model.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Hook + starting point** | Show the torch that won't work. Students answer the three ‘What do you already think?’ boxes. Collect ideas; do not correct yet. |
| 6–16 | **Build first** | Pairs build cell + switch + lamp from the picture, before any symbols. Ask: what did every working circuit have in common? Draw out ‘complete loop’. |
| 16–26 | **Picture → symbol → diagram** | Teach the solid row of symbols and the drawing rules. Model drawing the pairs' own circuit on the board with a ruler. |
| 26–32 | **Ammeter** | Add the ammeter in series; read it. Name ‘current’ and ‘series’ only. Charge, Q = I t and p.d. wait until Lessons 2–3. |
| 32–44 | **Try it** | Questions 1–5. Circulate on Q4 (spot the mistakes) and Q5 (draw it): these show whether the drawing rules have landed. |
| 44–50 | **Exam corner** | Q7 as a hinge: check symbols, loop and ammeter placement. Q6 (torch detective) for homework if needed. |

! **Hinge check.** Show circuit C from Q3. ‘Will the lamp light?’ If more than a few say yes, reteach ‘a gap anywhere stops the current everywhere’ before moving on.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| The cell ‘fills up’ the wire with electricity, which travels from the cell to the lamp. | Use the bike-chain model: the charges are already in the wire and all move together. |
| Current is used up by the lamp. | Plant the idea with two ammeters now; test it properly in Lesson 2. |
| Symbols can go anywhere near the wire. | Q4 makes the rule explicit: the wire must go into the symbol. |

## SUPPORT

Give out a symbol card set to match with real components before drawing. Allow tracing the symbol bank for Q2.

## STRETCH

Draw a circuit with two lamps and a switch that controls only one of them (early preview of parallel).

**Homework:** Workbook: Revise it, then check with the quick answers at the back.

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Lesson

# 02 CURRENT AND CHARGE

AQA Combined 6.2.1.2; Physics 4.2.1.2; Virtual Lab 1; 50 minutes

**BIG QUESTION** **Flick a switch and the lamp lights instantly, yet each electron drifts through the wire slower than a snail. How can both be true?**

**STUDENTS WILL BE ABLE TO**

- describe current as the rate of flow of charge
- use Q = I t, with time in seconds
- explain why the current is the same everywhere in a single loop

**WHY THIS LESSON, HERE**

Current is defined as a rate before students calculate with it. The model (bike chain, then ticket barrier) explains the instant switch-on puzzle and why current is conserved, so Q = I t is attached to meaning rather than a triangle.

## PREPARATION AND EQUIPMENT

Computers or tablets with PhET Circuit Construction Kit: DC (Lab). Virtual Labs worksheet pp. 3–6 for pairs who finish. Optional: a real series loop with two ammeters.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now** | Retrieval from Lesson 1. Then pose the big question and take predictions. |
| 6–18 | **Learn** | Free electrons, coulombs, current as a rate (ticket barrier). Slide 10: model both examples while students fill in the workbook frames, then each ‘Your turn’ alone. Quick check (slide 13) in the workbook box at the end of the lesson. |
| 18–32 | **Virtual Lab 1** | Pairs measure current at three points, then change R. Record in the Lab box. Ask: ‘Where did the current go?’ |
| 32–44 | **Try it** | Q1–Q5 in order: the calculation questions fade from full frame to no frame. Q6 is the challenge. Q7 is the misconception check. |
| 44–50 | **Exam corner** | Questions 10–11. Peer mark using the tip: time conversion, unit. |

! **Hinge check.** Q2 (0.60 A means…). Anyone choosing ‘0.60 C in the circuit’ needs the ticket-barrier model again.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| Current is used up. | Virtual Lab 1 plus Q7. Language: charge goes round; energy is transferred. |
| Minutes go straight into Q = I t. | Every supported frame has a ‘Convert’ row first. |
| Electrons flow from + to −. | Show both arrows on one diagram; exam diagrams use conventional current. |

## SUPPORT

Keep the frames; give a units card (C, A, s).

## STRETCH

How many electrons pass in 1 s at 1 A? (≈ $6 \times 10^{18}$.) Explain why a bigger current does not need faster-moving electrons in a thicker wire.

**Homework:** Prep 1 (Lessons 1–2). Finish Virtual Lab 1 pp. 4–6 if not done.

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Lesson 03

# POTENTIAL DIFFERENCE

AQA Combined 6.2.1.3, 6.2.4.2; Physics 4.2.1.3, 4.2.4.2; Virtual Lab 2; 50 minutes

**BIG QUESTION** **An AA battery says 1.5 V. A car battery says 12 V. What is that number actually telling you?**

### STUDENTS WILL BE ABLE TO

- explain p.d. as the energy transferred per coulomb
- connect a voltmeter in parallel across a component
- use E = Q V
- explain why the p.d.s in a loop add up to the supply p.d.

### WHY THIS LESSON, HERE

p.d. is the hardest idea in the unit. It is taught as energy per coulomb, with a model (delivery vans) and an explicit contrast with current, before it is used in V = I R. The voltmeter is introduced here, when students know what it measures.

## PREPARATION AND EQUIPMENT

PhET (Lab) and Virtual Labs worksheet pp. 7–10. Real demo: 6 V supply, two lamps in series, two voltmeters.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now** | Retrieval: current, Q = I t, current not used up. |
| 6–18 | **Learn** | Delivery-van model; 1 V = 1 J/C. Worked example and Your turn; slide 15 two-step example in its workbook frame. Voltmeter in parallel: say ‘across’. |
| 18–30 | **Virtual Lab 2** | Measure resistor p.d. at 3, 6, 9 V; calculate E for 2 C. Add a second resistor and test ‘p.d.s add up’. |
| 30–43 | **Try it** | Q1 (meter placement) as a class hinge first, then Q2–Q6. |
| 43–50 | **Exam corner** | Questions 9–10; model a two-idea explanation (joules + per coulomb). |

! **Hinge check.** Q1: which circuit measures the p.d. across the lamp? Students who pick B are still thinking of meters as ‘in the loop’.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| Voltage is the speed or ‘strength’ of the current. | Q6 (Leo). Energy per coulomb, not speed. |
| p.d. ‘at’ a point. | A p.d. is always between two points; insist on ‘across’. |
| Voltage is used up so current falls. | Energy is transferred; current stays the same (link to Lesson 2). |

## SUPPORT

Current vs p.d. comparison table on a card; sentence starter ‘___ J of energy is transferred for every ___’.

## STRETCH

A 9 V battery drives 0.50 A for 40 s. Find the charge and then the energy (two-step).

**Homework:** Revise it box, then check with the quick answers at the back.

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Lesson

# 04 RESISTANCE AND V = I R

AQA Combined 6.2.1.3–6.2.1.4; Physics 4.2.1.3–4.2.1.4; Virtual Lab 3; 50 minutes

**BIG QUESTION** **A dimmer switch makes a lamp glow less brightly. What is it actually changing?**

**STUDENTS WILL BE ABLE TO**

- describe resistance and its unit, the ohm (Ω)
- use V = I R, including mA and kΩ
- recognise an ohmic conductor from data or a graph
- explain why heating a metal increases its resistance

**WHY THIS LESSON, HERE**

V = I R now joins current and p.d., which students understand separately. The particle picture of resistance (collisions with ions) is taught here so that the lamp curve in Lesson 8 is a consequence, not a new fact.

## PREPARATION AND EQUIPMENT

PhET (Lab), Virtual Labs pp. 11–14. Demo: dimmer (variable resistor) with a lamp.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now** | Retrieval: p.d. meaning, voltmeter, p.d.s add up. |
| 6–18 | **Learn** | Resistance as opposition; two circuits at 6 V. V = I R: fill in the two slide 19 examples together, each with its Your turn, then both worked examples (the second includes mA). |
| 18–30 | **Virtual Lab 3** | Change battery only; V ÷ I stays at 10 Ω. Note the axis convention box. |
| 30–44 | **Try it** | Q1–Q4 (fading calculation support), Q5 table + graph, Q6 temperature. |
| 44–50 | **Exam corner** | Questions 9–10. |

! **Hinge check.** Q1: same battery, 10 Ω vs 30 Ω. Tests the direction of the relationship before any number work.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| Bigger resistance → bigger current (‘more resistor, more electricity’). | Two circuits side by side at the same p.d. |
| Any straight line means ohmic. | Straight line through the origin; check V ÷ I. |
| mA used as A. | Worked example 2 converts explicitly. |

## SUPPORT

Rearranging help: the equation triangles on the toolkit page (optional). Keep frames.

## STRETCH

Explain resistance in terms of electrons colliding with ions. Why does a lamp's resistance rise but a resistor's not (in normal use)?

**Homework:** Prep 2 (Lessons 3–4).

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Lesson

# 05 RESISTANCE OF A WIRE

AQA Combined RP 15; Physics RP 3; Part A; 50 minutes

**BIG QUESTION** **If you double the length of a wire, what happens to its resistance?**

**STUDENTS WILL BE ABLE TO**

- set up a circuit to find resistance from V and I
- identify and control the variables
- record, process and graph results
- write a conclusion using evidence

**WHY THIS LESSON, HERE**

The first required practical. Students can now draw the circuit, place both meters and calculate R, so the practical is about method, variables and evidence rather than new content.

## PREPARATION AND EQUIPMENT

Per group: low-voltage dc supply (≈ 2 V), switch, ammeter, voltmeter, 1.1 m of constantan/nichrome wire (e.g. 0.3 mm) taped to a metre ruler, two crocodile clips, leads.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now** | Retrieval: R = V ÷ I, heating, meter symbols. |
| 6–14 | **Plan** | Complete the variables table (Q1). Walk through the circuit and safety. Stress switch-off between readings. |
| 14–34 | **Practical** | Ten lengths, 10–100 cm. Readings in the Results table. Repeat any anomaly immediately. |
| 34–44 | **Process** | Calculate R, plot the graph, line of best fit. Graphs can be finished for homework. |
| 44–50 | **Analyse** | Q4 and Q5 (pattern and prediction). Exam corner Q9 can be set as homework. |

! **Hinge check.** Why switch off between readings? (Q7.) If students answer ‘to save the battery’, reteach heating → resistance.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| ‘Measure resistance with the voltmeter.’ | Resistance is calculated from V and I. |
| Line of best fit = dot-to-dot. | Model a single straight line through the origin region, ignoring the anomaly. |
| Increasing length reduces current, so ‘current is the DV’. | The DV is R (calculated). |

## SUPPORT

The length axis is pre-scaled; give a resistance scale (e.g. 2 cm for 1 Ω) and a partially completed example table.

## STRETCH

Q8: explain a non-zero intercept (zero error at the clip).

**Homework:** Finish graph. Exam corner six-mark method.

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Lesson

# 06 SERIES AND PARALLEL CIRCUITS

AQA Combined 6.2.2; Physics 4.2.2; 50 minutes

**BIG QUESTION** **When one bulb in your kitchen blows, why don't all the other lights in the house go out?**

**STUDENTS WILL BE ABLE TO**

- tell series and parallel circuits apart
- use the rules for current and p.d. in each
- calculate total resistance in series
- explain why adding a parallel branch lowers the total resistance

**WHY THIS LESSON, HERE**

The series/parallel rules are taught as consequences of the Lesson 2–3 ideas: current is conserved (so it splits and rejoins) and energy per coulomb is shared (so p.d.s add in series).

## PREPARATION AND EQUIPMENT

Per pair: cells, three identical lamps, two resistors, ammeters and voltmeters; or PhET. Mini whiteboards.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now** | Retrieval: V = I R, wire length, meaning of volt. |
| 6–18 | **Learn** | Build two lamps in series, then in parallel; unscrew one lamp in each. Fill the rules table from observation. |
| 18–28 | **Measure** | Measure the numbers shown in the two labelled circuits. Check 'adds up' each time. |
| 28–44 | **Try it** | Worked example + Your turn, then the slide 30 example in its workbook frame, then Q1–Q6. |
| 44–50 | **Exam corner** | Question 7. |

! **Hinge check.** Q5: 6 Ω and 12 Ω in parallel. Anyone choosing 18 Ω is adding in parallel.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| Current splits equally at every junction. | Use unequal branch currents (0.25 A, 0.15 A). |
| p.d. splits between parallel branches. | Each branch has the full supply p.d. |
| Adding a resistor always increases total resistance. | Checkout model: another route lowers it. |

## SUPPORT

Rules table on a card beside every question.

## STRETCH

Go further: $1/R = 1/R_1 + 1/R_2$ for two resistors in parallel (beyond AQA; label it as such).

**Homework:** Prep 3 (Lessons 5–6).

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Lesson

# 07 RESISTORS IN SERIES AND PARALLEL

AQA Combined RP 15; Physics RP 3; Part B; 50 minutes

**BIG QUESTION** **Two identical resistors, three ways to connect them. Which arrangement lets the most current flow?**

**STUDENTS WILL BE ABLE TO**

- measure the total resistance of resistors in series and in parallel
- compare results with $R_{total} = R_1 + R_2$
- explain the results using paths and current
- evaluate the measurements

**WHY THIS LESSON, HERE**

The second part of the resistance practical tests the Lesson 6 rules with measurements, and gives evaluation practice (specific error and improvement).

## PREPARATION AND EQUIPMENT

Per group: low-voltage supply, switch, ammeter, voltmeter, two identical resistors (e.g. 10 Ω), leads.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now** | Retrieval: series total, parallel rules. |
| 6–12 | **Predict** | Q1: rank the three arrangements. Record predictions on the board. |
| 12–32 | **Practical** | Four arrangements; V across the combination, I from the supply. |
| 32–44 | **Analyse** | Q3–Q6. Compare with predictions. |
| 44–50 | **Exam corner** | Question 7. |

! **Hinge check.** Before building: where does the voltmeter go for the parallel pair? (Across both.)

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| Measure V across one resistor only in parallel. | Across the combination. |
| Evaluation: ‘human error’. | Name a specific error (lead resistance, resolution, heating) and a matching improvement. |

## SUPPORT

Circuit diagrams printed at each station.

## STRETCH

Predict the result for three identical resistors in parallel and test it.

**Homework:** Workbook Revise it, then check with the quick answers at the back.

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Lesson

# 08 I–V GRAPHS

AQA Combined 6.2.1.4; Physics 4.2.1.4; 50 minutes

**BIG QUESTION** **Every component has a ‘fingerprint’: the shape of its I–V graph. Could you identify a component from its graph alone?**

### STUDENTS WILL BE ABLE TO

- read an I–V graph, including negative values
- sketch and recognise the graphs for a fixed resistor, a filament lamp and a diode
- calculate the resistance at a point with R = V ÷ I
- explain the shape of each graph

### WHY THIS LESSON, HERE

I–V graphs are taught as ‘fingerprints’ after students already know R = V ÷ I, ohmic behaviour and why heating raises resistance. The graph shapes follow from ideas they already have.

## PREPARATION AND EQUIPMENT

Visualiser. Optional demo: lamp + variable supply with meters, to show resistance rising. Mini whiteboards for sketching.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now** | Retrieval: ohmic, heating, R at a point. |
| 6–20 | **Learn** | Reading an I–V graph (both quadrants). Build the three shapes one at a time; students sketch each. |
| 20–26 | **Explain** | Lamp chain: current → heating → vibrating ions → collisions → R up → less steep. |
| 26–44 | **Try it** | Q1–Q5. Q3 (card-sort order) makes the causal chain explicit. |
| 44–50 | **Exam corner** | Question 6: sketch plus explanation. |

! **Hinge check.** Q1: identify the three unlabelled graphs.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| Resistance = gradient of the I–V graph. | Use R = V ÷ I at a point; a curve's gradient does not give R. |
| The lamp graph flattens because the current ‘runs out’. | Heating raises resistance. |
| Diode conducts both ways but less one way. | Very high resistance in reverse: almost zero current. |

## SUPPORT

Sketch templates with axes pre-drawn.

## STRETCH

Why does a diode pass almost no current until the p.d. reaches about 0.6 V?

**Homework:** Prep 4 (Lessons 7–8).

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Lesson

# 09 INVESTIGATING I–V CHARACTERISTICS

AQA Combined RP 16; Physics RP 4; 50 minutes

**BIG QUESTION** **Can you produce the fingerprint graphs of a resistor, a lamp and a diode yourself?**

**STUDENTS WILL BE ABLE TO**

- set up a circuit to vary the p.d. and measure the current
- take readings in both directions, safely
- plot an I–V graph with positive and negative values
- describe how the method changes for a diode

**WHY THIS LESSON, HERE**

Students now know what each graph should look like, so the practical tests predictions and focuses on method: varying p.d., reversing, safety and the diode's extra steps.

## PREPARATION AND EQUIPMENT

Per group: variable low-voltage supply or supply + variable resistor, ammeter (and milliammeter), voltmeter, fixed resistor, 6 V filament lamp, diode + protective resistor (e.g. 100 Ω), leads.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now + safety** | Retrieval. Lamp rating, hot lamp, protective resistor. |
| 6–12 | **Set up** | Check ammeter in series and voltmeter across the component only. |
| 12–34 | **Practical** | Resistor and lamp in both directions; diode with small steps. |
| 34–44 | **Graph** | Both components on one centred graph; sketch the diode. |
| 44–50 | **Analyse** | Q4–Q8; exam corner six-mark method as homework. |

! **Hinge check.** How do we get the negative readings? (Swap the supply connections; record as negative.)

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK... | ADDRESS IT BY... |
|---|---|
| Plotting reversed readings as positive. | Q8. |
| Voltmeter across the diode and protective resistor. | Across the diode only. |

## SUPPORT

Given target p.d.s in the table.

## STRETCH

Explain why the lamp's resistance at 1 V is still larger than at 0.1 V.

**Homework:** Exam corner six-mark question.

Lesson

# 10 THERMISTORS, LDRS AND SENSORS

AQA Combined 6.2.1.4, 6.2.2; Physics 4.2.1.4, 4.2.2; 50 minutes

**BIG QUESTION** **How does a streetlight know it's getting dark, and how does a fire alarm know the room is hot?**

**STUDENTS WILL BE ABLE TO**

- describe how a thermistor's resistance changes with temperature
- describe how an LDR's resistance changes with light intensity
- predict the current in a sensor circuit
- choose the right sensor for a job

**WHY THIS LESSON, HERE**

Thermistors and LDRs are taught as applications of 'lower total R → bigger current at fixed p.d.', so they revise V = I R and series circuits while adding two new components.

## PREPARATION AND EQUIPMENT

Thermistor and LDR with multimeter (resistance range); beaker of warm water; torch; mini whiteboards.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now** | Retrieval: lamp graph, V = I R direction, diode symbol. |
| 6–18 | **Learn + demo** | Measure R of the LDR (cover/torch) and thermistor (hand/warm water). Sketch the two graphs. |
| 18–28 | **Sensor circuits** | LDR in series; worked example and Your turn. |
| 28–44 | **Try it** | Q1–Q5. |
| 44–50 | **Exam corner** | Question 6. |

! **Hinge check.** Q5: which change increases the current through a thermistor?

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| Brighter light increases LDR resistance. | Both are 'more → less'. |
| Changing resistance changes the supply p.d. | The supply p.d. is fixed; the current changes. |

## SUPPORT

The 'more → less' key box as a card.

## STRETCH

Explain why the p.d. across the fixed resistor rises as the LDR is lit (qualitative potential divider).

**Homework:** Prep 5 (Lessons 9–10).

Lesson

# 11 POWER AND ENERGY TRANSFER

AQA Combined 6.2.4.1–6.2.4.2; Physics 4.2.4.1–4.2.4.2; 50 minutes

**BIG QUESTION** **Which transfers more energy: a 3 kW kettle boiling for 3 minutes, or a 10 W LED lamp left on for 5 hours?**

**STUDENTS WILL BE ABLE TO**

- describe power as the energy transferred per second
- use E = P t, P = V I and P = I² R with the right units
- describe the energy transfers in everyday appliances
- choose the right equation for a question

**WHY THIS LESSON, HERE**

Power pulls together earlier quantities. Students choose between equations (the decision table) instead of memorising a list, and P = I² R prepares the National Grid argument in Lesson 12.

## PREPARATION AND EQUIPMENT

Appliance rating plates (photos or real). Calculators.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now** | Retrieval: E = Q V, Q = I t, V = I R. |
| 6–18 | **Learn** | Power as J per s; appliance table; E = P t: slide 43 example in its workbook frame (unit conversions). |
| 18–28 | **More power equations** | P = V I and P = I² R: the slide 42 examples in their workbook frames; ‘square the current first’. |
| 28–44 | **Try it** | Q1 is the big question; then Q2–Q6. |
| 44–50 | **Exam corner** | Question 7 (six marks). |

! **Hinge check.** Which equation? Show three sets of data and ask students to name the equation only.

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| P = I² R calculated as (I R)² or I × 2 × R. | Square first, then multiply. |
| kW and minutes substituted directly. | Convert row in every frame. |

## SUPPORT

Decision table on a card.

## STRETCH

Q6 (lamp current and resistance); cost of energy if the teacher adds kWh.

**Homework:** Workbook Revise it, then check with the quick answers at the back.

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Lesson

# 12 MAINS, SAFETY AND THE NATIONAL GRID

AQA Combined 6.2.3, 6.2.4.3; Physics 4.2.3, 4.2.4.3; 50 minutes

**BIG QUESTION** **Pylon cables carry up to 400 000 V. Why so high, and why do we only get 230 V at the plug?**

**STUDENTS WILL BE ABLE TO**

- describe the difference between ac and dc
- state the UK mains values: 230 V, 50 Hz
- explain how the earth wire and fuse protect you
- explain why the National Grid uses very high p.d.s

**WHY THIS LESSON, HERE**

Mains safety and the National Grid close the unit as one story, from power station to plug. The grid explanation uses P = V I and P = I² R from Lesson 11.

## PREPARATION AND EQUIPMENT

Cut-away three-core cable or plug (no mains work by students). Transformer demo if available.

## LESSON SEQUENCE

| MIN | PHASE | TEACHER MOVES (WORKBOOK QUESTION NUMBERS IN THE LESSON) |
|---|---|---|
| 0–6 | **Do now** | Retrieval: P = V I, P = I² R, units. |
| 6–16 | **ac/dc and mains** | Traces or diagrams; 230 V, 50 Hz; cable colours and jobs. |
| 16–26 | **Safety story** | Live danger; earth + fuse sequence. Students order the events (Q3). |
| 26–36 | **National Grid** | Grid diagram; Q5 calculation shows the ×10 / ×100 effect. |
| 36–50 | **Try it + exam corner** | Q1–Q6, then exam corner Questions 7–8. Set the Unit review for home. |

! **Hinge check.** Q4 (Ava): is the live wire dangerous when switched off?

## MISCONCEPTIONS TO EXPECT

| STUDENTS MAY THINK… | ADDRESS IT BY… |
|---|---|
| Earth wire carries current all the time. | Only during a fault. |
| Step-up transformers make extra energy. | Power in = power out; p.d. up, current down. |
| Neutral is ‘safe’ and live only matters when on. | Live is at ~230 V whenever connected. |

## SUPPORT

Fault story as cut-out cards.

## STRETCH

Higher-tier transformer question (Q6); explain why ac is needed for transformers.

**Homework:** Prep 6 (Lessons 11–12), then the Unit review booklet.

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